

Robust Passivity and Passivity Relaxation for Impedance Control of Flexible-Joint Robots with Inner-Loop Torque Control

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Abstract—Passivity is a canonical condition for the safety of interactive systems, but practical limitations restrict its utility as a design tool. A system with a passive model can be unstable in high-stiffness environments, passivity is difficult to show with inner-loop controllers, and as it is a binary condition it provides limited design comparison insight; as a result, it is rarely used for inner-loop design. As passivity safety claims are limited by model accuracy, conditions for the passivity of a system with bounded-magnitude model uncertainty (robust passivity) are developed in this paper. Additionally, a condition for coupled environment–robot stability is developed using mixed passivity and small-gain condition, allowing rigorous relaxation of passivity at high frequencies for typical impedance-controlled systems. These approaches are used in the analysis of an impedance-controlled series-elastic actuated system with inner-loop torque control and also compared with traditional design tools (bandwidth ratio, sensitivity function, etc.). The approach is then validated experimentally, identifying model uncertainty bounds under various load conditions, and then using the measured uncertainty for controller synthesis. Robust passivity is then compared with nominal passivity in a validation experiment under manual excitation and impact.

Index Terms—Linear feedback control systems, robust control, torque control.

I. INTRODUCTION

INTERACTIVE robots regulate the apparent dynamics between external force and position [1], [2], allowing explicit design of robot behavior in uncertain environments. To accommodate external forces instead of rejecting them requires shifting actuator design goals toward backdriveability and compliance. Mechatronic design for interactive robots must balance performance and robustness [3], but traditional design tools (e.g., stability robustness, disturbance rejection, and gain/phase

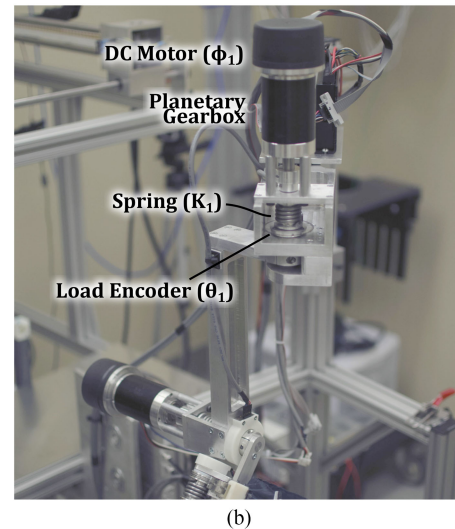
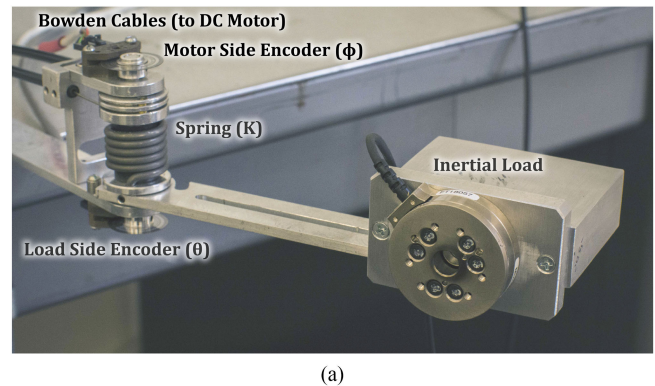


Fig. 1. Series-elastic systems, with both linear and nonlinear load side dynamics. (a) Single joint test setup used for validation. (b) First joint of a five DOF upper-limb exoskeleton.

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margin) that classically give an insight into this tradeoff must be reconsidered for interactivity. Practical safety conditions are needed to make a system's limitations explicit and understand the impact of mechatronic design on these limitations.

The dominant approach to demonstrate interactive robot safety is passivity, which offers rigorous stability guarantees for the interconnection of a passive robot with a passive environment. Once shown analytically, passivity should provide universal safety guarantees, but real systems with analytically passive models can still be unstable when coupled with some

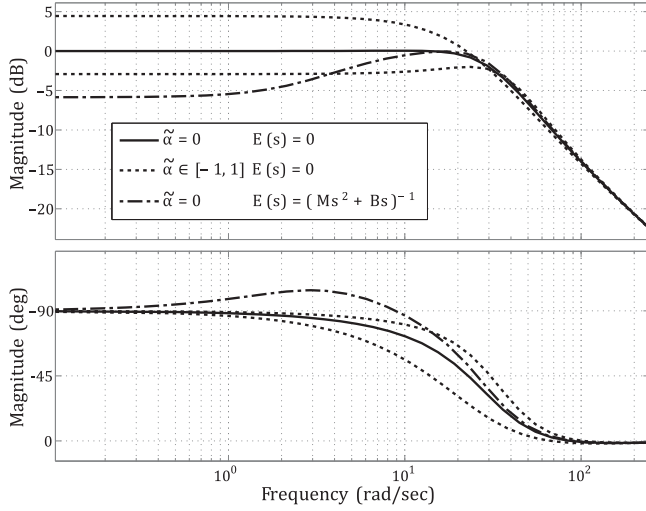


Fig. 3. Effect of effective backdriveability ($\hat{\alpha}$) and load dynamics $E(s)$ on torque tracking $\tau^d \rightarrow \tau$ for a PD torque controller.

$G(s) = (Ms^2 + Bs)^{-1}$, proportional derivative (PD) torque controller $C(s) = K_d s + K_p$, the effect of environment dynamics is shown in Fig. 3, where the torque response of a fixed output [$E(s) = 0$] and a damped inertia load [$E(s) = 0.3s^2 + 0.5s$] are compared. Low-frequency magnitude gain is lost when environment admittance increases. Advanced approaches in force control [29] can compensate environment dynamics, but require an accurate dynamic model making them unsuitable for interactive robots.

This sensitivity to environment dynamics is undesirable when abstraction of the robot to the desired impedance is necessary, such as when analytic or iterative techniques are used to choose impedance parameters. The loss in inner-loop bandwidth makes the hierarchical design rule of thumb, that the inner loop can be ignored when its bandwidth is 5–10 times that of the outer loop, environment specific. However, for other applications, impedance parameters are experimentally tuned for each environment, making changes in torque loop performance due to environment dynamics less important.

C. Backdriveability

The “reference” for interactive robots enters the system physically (as opposed to the typical digital entry for most control systems). This involves the physical backdriveability of the drivetrain, which also impacts torque-loop performance but can be difficult to model. As the gearbox carries higher torques, reaction forces on gear supports increase, changing friction characteristics. Drivetrains, which are considered backdriveable [9], [15] and nonbackdriveable [20], [30], have been investigated, but no consensus on best practice has been reached.

Here, a linearized backdriveability will be considered where the load torque seen at the motor is $\alpha\tau$. For nonbackdriveable systems $\alpha = 0$, and fully backdriveable drivetrains have $\alpha = 1$. Commonly, backdriveable systems will include a feedforward term in the controller $\tau_m = C(s)(\tau^d - \tau) + \hat{\alpha}\tau$, where $\hat{\alpha}$ is the estimated backdriveability [9]. However, uncertainty in spring

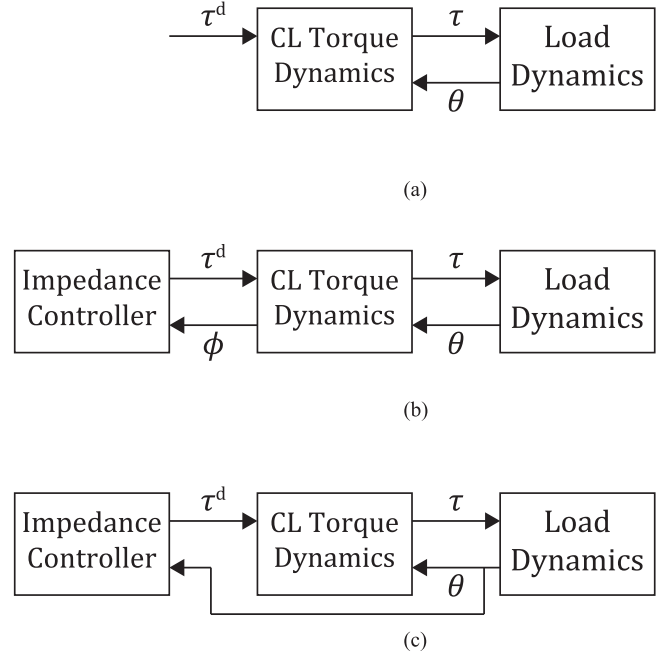


Fig. 4. Structure of passivity arguments for impedance control systems with closed-loop torque dynamics. (a) Inner-loop torque passivity. (b) Motor-side impedance control. (c) Load-side impedance control.

stiffness, backlash in spring mounting and gearbox, and other nonlinearities cause a difference between effective α and $\hat{\alpha}$. The effect of backdriveability on torque-tracking performance can be seen in Fig. 3, where $\tilde{\alpha} = \alpha - \hat{\alpha}$ is the effective backdriveability remaining after compensation. If backdriveability is under compensated, $\tilde{\alpha} > 0$ and low-frequency magnitude response is reduced (which can be observed on some torque-tracking responses seen in the literature). If $\tilde{\alpha} < 0$, the overcompensated backdriveability phase increases in the neighborhood of the roll-off frequency and passivity can be more easily lost. The effect of this uncertainty is reduced under high-gain control, but the underlying uncertainty remains, potentially compromising safety or stability guarantees.

III. ROBUST PASSIVITY

Passivity stands as a fundamental and intuitive safety condition and is introduced in this section as it is relevant for impedance-controlled systems.

A. Nominal Passivity

Prior work using passivity for impedance control of flexible-joint robots will be introduced.

1) *Torque Control Passivity*: The nominal passivity of an inner-loop torque controller as seen in Fig. 4(a) has been shown to induce lower bounds on the parameters of a fixed-structure controller (PD), but assumes τ_d to be the exogenous input [14]. As the passivity of systems in series cannot be concluded from the passivity of each individual system, this approach does not extend to when there is an outer-loop impedance controller.

2) *Motor Side Impedance Passivity*: One approach is to realize impedance control on the motor position as seen in Fig. 4(b)

[11]. A desired load position is translated to a motor-side position which, in steady state, induces the desired load-side position. As each coupling is power continuous, overall passivity can be directly concluded from individual system passivity. However, as joint stiffness decreases (e.g., with SEAs, variable stiffness actuators, etc.) the accuracy of rendered impedance in this approach is reduced —apparent dynamics present the physical stiffness in series with the desired impedance.

3) Load Side Impedance Passivity: Taking the impedance rendered from load-side position as shown in Fig. 4(c), under the controller architecture seen in Fig. 2 with $\alpha = 0$, the total passivity condition is

$$\frac{\tau}{\dot{\theta}} = \frac{I(s) \cdot KG(s)C(s) - K}{s + sKG(s)C(s)} \in \mathcal{P} \quad (2)$$

where $\mathcal{P}$ denotes the set of passive systems. Note that passivity applies to velocity/torque relations (where power is their product).

B. Passivity With Model Uncertainty

Conditions for the passivity of a system with bounded-magnitude multiplicative uncertainty (robust passivity) will be developed.

1) Uncertainty Representation: Let a multiplicative uncertainty term $\Delta(\omega)$ relate real system G and model $\hat{G}$ as

$$G(j\omega) = \hat{G}(j\omega)(1 + \Delta(\omega)), \quad |\Delta(\omega)| < \overline{\Delta}(\omega). \quad (3)$$

The uncertainty has an unknown phase and a frequency-dependent magnitude bound, allowing the description of unmodeled dynamics such as structural resonance, electrical dynamics, or backdriveability.

2) Passivity of General Uncertain System: Passivity of a linear system is most easily represented as a phase-based condition, but introducing a term with an unknown phase limits this approach. The Cayley transform [31] allows the transformation of passivity to a small-gain condition on a related system. If Q is a square, causal system the following equivalence holds:

$$Q \in \mathcal{P}^+ \iff \left\| \frac{Q - 1}{Q + 1} \right\| < 1 \quad (4)$$

where $\mathcal{P}^+$ is the set of strictly passive systems (strictly input passive), $\|\cdot\|$ is an admissible system norm, and 1 is extended to the identity matrix for multi-input, multi-output systems. Rewriting (4) for a linear, time invariant, single-input, single-output $Q(s)$ and using the $\mathcal{H}_\infty$ norm gives

$$\left| \frac{A(j\omega)\Delta(\omega) + N(j\omega)}{B(j\omega)\Delta(\omega) + M(j\omega)} \right| < 1 \quad \forall \omega. \quad (5)$$

For a given system and controller which meet nominal robustness, a robust passivity bound $\overline{\Delta}_p(\omega)$ that guarantees passivity for all $|\Delta(\omega)| < \overline{\Delta}_p(\omega)$ can be found computationally over a set of frequencies Ω with Algorithm 1.

3) Robust Passivity of Impedance Control: Using the Cayley transform to extend the passivity condition (2) to hold with

Algorithm 1: Uncertainty Bound Fitting.

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1: for  $\omega \in \Omega$  do
2:   while  $\exists \xi$  s.t.  $|A\overline{\Delta}_p(\omega)e^{(\xi i)} + N| > |B\overline{\Delta}_p(\omega)e^{(\xi i)} + M|$  do
3:      $\overline{\Delta}_p(\omega) \leftarrow \gamma \overline{\Delta}_p(\omega), \quad 0 < \gamma < 1$ 
4:   end while
5: end for

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model uncertainty (3) gives

$$\left| \frac{(\hat{G}C(I - j\omega))(1 + \Delta(\omega)) - 1 - K^{-1}j\omega}{(\hat{G}C(I + j\omega))(1 + \Delta(\omega)) - 1 + K^{-1}j\omega} \right| < 1 \quad \forall \omega. \quad (6)$$

This can be readily rewritten in the form (5). Once this condition is satisfied for given $I_0, C, \hat{G}$, the rendered impedance I can be scaled $I = \eta I_0$, $\eta \in [0, 1]$ while maintaining robust passivity, which can be directly shown with the time-domain definition of passivity.

C. Frequency-Domain Passivity Relaxation

If the environment is linear, mixed passivity and small-gain condition can be used to conclude coupled stability [32], which has been explored for controller synthesis of flexible-link manipulators [33], but not interactive robots.

Denoting the environmental dynamics $\tau \rightarrow \dot{\theta}$ as $E_v(s)$ and rendered robot dynamics as $H(s)$ [here, (2)] and assuming $E_v(s)$ and $H(s)$ to be stable, square, and strictly proper linear time-invariant (LTI) systems, divide the frequency domain into two sets, Ω_p and Ω_s such that $\Omega_p \cup \Omega_s = (-\infty, \infty)$. Let the inner product $\langle g, f \rangle_\Omega = \frac{1}{2\pi} \int_{\omega \in \Omega} f^*(j\omega)g(j\omega)d\omega$. For an LTI, stable, square, strictly proper system G with input u and output y , frequency limited strictly input passivity and small-gain properties are as follows:

$$\exists \delta > 0 \text{ s.t. } \langle u, Gu \rangle_{\Omega_p} \geq \delta \langle u, u \rangle_{\Omega_p} \quad \forall u \quad (7)$$

$$\langle Gu, Gu \rangle_{\Omega_s} \leq k_2^2 \langle u, u \rangle_{\Omega_s} \quad \forall u. \quad (8)$$

By Th. 1 of [32], if H and E_v are strictly passive (strictly input passive) over Ω_p and small gain over Ω_s ($k_2 < 1$), the coupled system is input–output stable.¹

Here, it is desired to use a $\mathcal{H}_\infty$ norm, and noting

$$\langle Gu, Gu \rangle_{\Omega_s} \leq k_\infty^2 \langle u, u \rangle_{\Omega_s} \quad (9)$$

$$k_\infty = \max_{\omega \in \Omega_s} |G(j\omega)| \quad (10)$$

an $\mathcal{H}_\infty$ gain $k_\infty < 1$ is sufficient for (8). Let $\Omega_p = \{\omega : |\omega| < \omega_0\}$ and $\Omega_s = \{\omega : |\omega| > \omega_0\}$. Then, let ω_0 be such that

$$\min_{\omega_0} \text{ s.t. } \forall \omega \geq \omega_0, \quad |H(j\omega)||E_v(j\omega)| < 1 \quad (11)$$

satisfies (8) for H and E_v , allowing the scaling of the two systems while preserving the loop gain.

¹ See the discussion at the end of Section IV in [32] for relaxing the very strict passivity condition of Theorem 1 to strictly input passive.

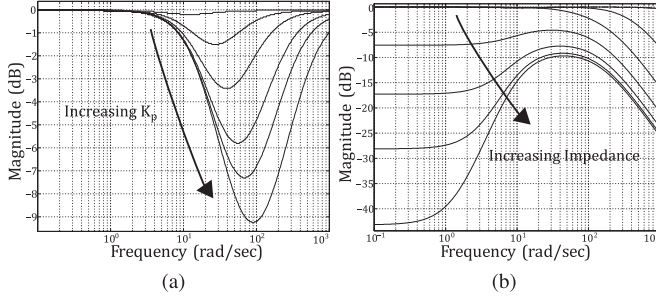


Fig. 5. Allowed uncertainty $\bar{\Delta}_p(j\omega)$ which guarantees robust passivity as (a) inner-loop proportional gain is changed and (b) outer-loop impedance is scaled with fixed zero location.

To state as relevant for impedance-controlled robots: H and E_v are coupled input–output stable if there exists ω_0 , which meets (11), E_v is strictly input passive and H meets

$$\left| \frac{H(j\omega) - 1}{H(j\omega) + 1} \right| < 1 \quad \forall |\omega| < \omega_0. \quad (12)$$

As the passivity constraint on $H(j\omega)$ need hold only up to ω_0 , this is referred to as a relaxed passivity condition and can be used to relax nominal passivity as well as robust passivity, such as in (6).

IV. ANALYSIS

In this section, the proposed approaches (passivity robustness and passivity relaxation) will be examined as potential design tools.

A. PD Torque Control

One advantage of robust passivity is that the passivity robustness of different designs can be compared by producing an allowable uncertainty $\bar{\Delta}_p(\omega)$. Unless otherwise noted, consider a PD inner-loop controller with a stiffness/damping impedance controller ($C(s) = K_p + K_d s$, $I(s) = K_{\text{imp}} + B_{\text{imp}} s$ in the configuration of Fig. 2).

1) **Robust Passivity Bound $\bar{\Delta}_p(\omega)$:** Using Algorithm 1 to find $\bar{\Delta}_p(\omega)$ for several sets of parameters, the effect of two controller changes are found as seen in Fig. 5(a) and (b). Increasing the inner-loop proportional gain K_p decreases the passivity robustness. Increasing impedance by scaling I (with impedance zero fixed, $K_{\text{imp}}/B_{\text{imp}} = 10$) also decreases the passivity robustness. These two trends show the expected effect of high-gain controllers—decreasing robustness. However, the frequency profile of these changes is different—as models are more accurate at low frequencies, low-frequency robustness is easier to satisfy, requiring interpretation of changes in the shape of $\bar{\Delta}_p(\omega)$.

The individual scaling of model and controller parameters shows the impact of each parameter on passivity robustness. For compactness, robustness will be characterized as the minimum of $\bar{\Delta}_p$ over the range $\omega \in \{10^{-2}, 10^3\}$ rad/s. The results can be seen in Table I, and note that allowable uncertainty of $-\infty$ implies that even nominal passivity is lost.

Increasing the motor damping B or the inner-loop derivative K_d unilaterally improves robustness. This matches prior work, which shows that increasing damping (physical and from the controller) has a positive impact on safety for interactive systems [4]. Increasing motor inertia I , inner-loop proportional gain K_p , and outer-loop stiffness K_{imp} unilaterally decrease the robustness. The latter two are controller gains, matching the established intuition that high control gain decreases robustness. Increasing motor inertia makes inner-loop dynamics more significant at low frequencies (i.e., decreases inner-loop bandwidth), contributing a phase that challenges even nominal passivity.

Other parameters do not have a uniform impact on robustness. Increasing B_{imp} should improve passivity as it dissipates more power, but robustness decreases at high frequencies for large values of B_{imp} . One explanation for this is that increasing B_{imp} amplifies the high-frequency input to the inner loop, thereby increasing the contribution of high-frequency inner-loop dynamics. Very low stiffnesses K present poor robustness, but it does not significantly change between higher stiffnesses—the impact of joint stiffness in this plateau can be better understood with passivity relaxation.

2) **Passivity Relaxation:** Real, controlled interactive systems present passivity violations at high frequencies due to higher order dynamics (e.g., link flexibility), electrical dynamics, or discretization effects. At the same time, impedance-controlled robots roll-off at high frequencies, where the actuator position response is minimal, and high-frequency sinusoidal velocity input does not develop output torque. The environment presented to real impedance-controlled robots also rolls-off at high frequencies, as there is at least the inertia of the mechanism, which connects the elastic element to the environment (although this roll-off may be impractically high in some systems).

A stiffness environment requires a higher ω_0 than a pure inertial environment, but it does not change the controlled system’s ability to meet (12), making passivity relaxation more difficult to meet. This provides an explanation to the fact that while the theoretical “worst-case environment” could be a pure stiffness or a pure inertia [34]; in practice, a pure stiffness environment is more challenging for interactive robots.

Take the rendered robot dynamics (2) as H , rendering a high-frequency impedance of K/s . Increasing joint stiffness K thus increases the high-frequency magnitude of H , increasing ω_0 for a fixed environment. However, increasing K will also impact the stability condition (11), and a full analysis is outside the scope of this paper.

Assume a second-order environment as $E_v = s(M_e s^2 + B_e s + K_e)^{-1}$. An impedance robot interacting with a kinematic constraint (i.e., large K_e) also has M_e of the link inertia of the robot but limited damping, giving an under-damped resonance. If passivity cannot hold beyond this resonance frequency (roughly $\sqrt{K_e/M_e}$), the robot and the environment must meet the small-gain condition for relaxed passivity. The closed-form expression of the maximum magnitude of a second-order system can be written and simplified by letting $B_e^2/4 \rightarrow 0$, and

TABLE I
EFFECT OF PARAMETERS ON ROBUST PASSIVITY, ROBUSTNESS REPORTED AS $\min_{\omega \in \{10^{-2}, 10^3\}} |\bar{\Delta}_p(\omega)|_{dB}$

Parameter:	Nominal:	Scaling Factor:	0.01	0.1	1	10	100	$K_i =$	0	0.25	0.5	0.75	1
I	0.032 [Kg·m ²]	$\min \bar{\Delta}_p :$	-0.5	-2.5	-9.4	-18.9	-28.8	$\min \bar{\Delta}_p :$	-9.4	-9.5	-9.7	-9.7	-9.8
B	5.8e-4 [Nm·s/rad]		-9.4	-9.4	-9.4	-8.9	-6.0	$K_v =$	0.01	0.05	0.1	0.5	1
K	16.4 [Nm/rad]		-inf	-120.3	-9.4	-9.6	-9.8		-7.1	-5.0	-3.8	-4.5	-7.2
K_p	5		-5.8	-8.8	-9.4	-50.6	-inf	$\bar{\alpha} =$	-1	-0.5	0	0.5	1
K_d	0.1		-inf	-21.2	-9.4	-2.4	-0.4		-14.0	-11.2	-9.4	-9.1	-8.9
K_{imp}	1		-9.4	-9.4	-9.4	-13.0	-inf	BW ratio =	1.5	2	3.4	5	10
B_{imp}	0.01		-25.0	-18.2	-9.4	-19.9	-66.5		-20.9	-18.4	-14.8	-12.1	-7.2

(a) Nominal parameter values, and their impact on passivity robustness.

(b) Impact of various controller architectures. All parameters from Table I(a) remain nominal.

assuming $H(s) = K/s$ as

$$K \cdot \frac{\sqrt{M_e}}{B_e \sqrt{K_e}} < 1. \quad (13)$$

This condition, which allows passivity violations at environmental resonance, shows a tradeoff between environmental stiffness and joint stiffness, where high-stiffness environments can provide coupled stability guarantees with a lower joint stiffness.

B. Alternative Inner-Loop Controllers

Explicit torque control is a common choice for inner-loop control on flexible joint robots, in both PID [22], [35] and PD [20], [36] varieties. For PID torque control, integral action shows little impact on robustness compared to PD, as can be seen in Table I. Disturbance observer approaches are also common, and though tractable within this framework, require analysis outside the scope of this paper.

Position control is often proposed for SEAs, where a position controller C_p is used, with motor torque expression $\tau_m = C_p(s)(\phi^d - \phi)$, and motor reference ϕ^d chosen to induce desired torque $\phi^d = \theta + K^{-1}\tau^d$. This can be rewritten as explicit torque control $C(s) = C_p(s)K$.

Torque control is also used with an innermost-loop velocity controller, giving three nested loops—impedance, torque, and velocity [14], [21], [30], [37]. Suppose a velocity controller of $\tau_m = K_v(\dot{\phi}^d - \dot{\phi})$ and torque control loop $\dot{\phi}^d = C(s)(\tau^d - \tau)$. This architecture decreases low-frequency robustness but improves moderate to high frequency robustness (>5 rad/s). The effect of varying velocity gain K_v seen in Table I is not unilateral, and robustness peaks at moderate values.

Feedforward terms can also be introduced to improve torque-control performance. As discussed in Section II-C, a feedforward torque term ($\hat{\alpha}\tau$) can help improve lower-frequency torque performance on backdriveable systems. However, overcompensation decreases effective backdriveability $\bar{\alpha}$, decreasing $\bar{\Delta}_p$ as seen in Table I. Velocity feedforward terms, i.e., viscous friction compensation, can be dangerous for interactive robots as they directly inject power to the system. Interpreting velocity feedforward as reducing the effective viscous damping in motor dynamics B , this decreases the robustness as seen in Table I.

C. Bandwidth Ratio Comparison

One motivation for avoiding passivity arguments in inner-loop design is the argument that inner-loop dynamics can be

ignored when inner-loop bandwidth is 5–10 times the outer-loop bandwidth. Originally used in the context of tracking control, applying this to interactive control requires interpreting the impedance controller as a tracking controller ($\theta^d \rightarrow \theta$), an expression which depends significantly on environment dynamics. Counter-examples can be constructed where a system fails passivity but has a bandwidth ratio exceeding ten by setting low derivative terms on the inner-loop controller. However, for more typical configurations ($K_p/K_d \approx 5$), robustness increases with bandwidth ratio. In Table I, the outer loop bandwidth is found, assuming a moderate admittance environment $E(s) = (.3s^2 + .1s + 10)^{-1}$ and the bandwidth ratio is varied by scaling the inner-loop controller. Passivity robustness increases with bandwidth ratio, but has no rigorous connection to passivity and does not capture the importance of the inner-loop derivative action.

D. Sensitivity and Complementary Transfer Functions

Note that the equation for total system response of Fig. 2 can be rewritten in terms of the sensitivity $S(s) = (1 + KGC)^{-1}$ and complementary sensitivity $T(s) = KGC(1 + KGC)^{-1}$ functions as

$$\frac{\tau}{\theta} = I(s)T(s) - KS(s). \quad (14)$$

Traditional interpretations of sensitivity S (i.e., in tracking control) is the system's sensitivity to external disturbance. Here it is also the torque response of the physical spring, causing a deviation from the desired impedance response. Thus, the classical design goal of minimizing $|S|$ is inherited, making Bode's integral relevant as

$$\int \log |S(j\omega)| d\omega = -\pi \sum p_k - \frac{\pi}{2} \lim_{s \rightarrow \infty} sL(s) \quad (15)$$

where p_k is right-half plane poles (here, none) and $L(s)$ is the loop-transfer function [here, $KG(s)C(s)$]. Also called the waterbed effect, reducing the sensitivity at some frequencies requires increasing it at others. With a damped inertia motor model and PD inner-loop controller, the integral constraint in (15) becomes $\int \log |S(j\omega)| d\omega \propto -K_d M^{-1}$. Increasing the inner-loop derivative gain K_d allows unpunished reductions in the sensitivity function, offering an improved performance, while increasing the motor inertia M reduces this allowance. This reinforces the impact of M and K_d on passivity seen in Table I—these

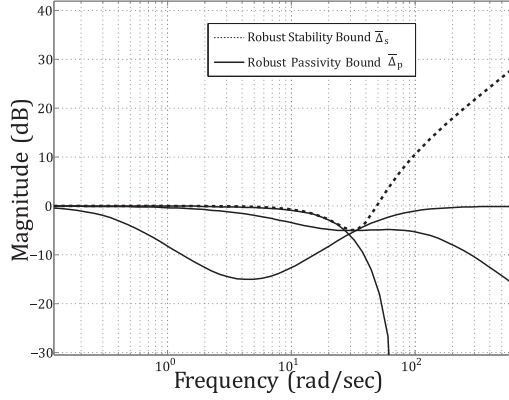


Fig. 6. Comparison of maximum allowable uncertainty magnitude for robust passivity to robust stability over various values of B_{imp} .

parameters have a significant impact on both performance and safety.

Complementary sensitivity function T characterizes how much uncertainty can be tolerated in a robust stability condition. Using a standard robust stability criteria [38] over multiplicative uncertainty (3) guarantees a robust internal stability of (14) for all $|\Delta(\omega)| < |T(j\omega)|^{-1} \forall \omega$. For a nominally stable system, the maximum magnitude of multiplicative uncertainty is found as $\bar{\Delta}_s(\omega) = |T(\omega)|^{-1}$, shown along with robust passivity $\bar{\Delta}_p$ in Fig. 6, where the impedance damping term B_{imp} is varied. Robust passivity is more conservative than robust stability, but the form of this conservatism depends on impedance zero location. Although intuitive that robust passivity is more conservative than robust stability, this reinforces that designs which are stable in isolation (i.e., meet robust stability in practice) can be unsafe in interaction.

V. EXPERIMENTAL VALIDATION

To validate the proposed approach, the 1-DOF system seen in Fig. 1 was used for identification and testing. A Maxon dc motor (EC-max 30), reduced by a 1:132 Maxon planetary gearbox is connected through Bowden cables to the motor side of a torsional spring (16.4 N·m/rad). The drivetrain has effective output inertia of 0.032 kg·m² and nominal output damping of 5.83e−4 (N·m·s/rad) [35]. Coulomb friction is identified from break-away values (0.12 N·m), and compensated in the following.

A. Uncertainty Bounds Identification

To identify the uncertainty in motor dynamics, the motor is identified in three configurations. In the first two configurations, the identification is driven by motor excitation where a series of sinusoidal current commands of varying magnitudes and frequencies was generated. The first three cycles are discarded, the remaining signal bandpass filtered, and a sinusoidal least-square fit used to find the magnitude and phase of the response. The frequencies and amplitudes were chosen experimentally to keep a measurable response and stay within torque and joint angle limits.

The first load condition removes the spring from the system and identifies from motor current to achieved motor position

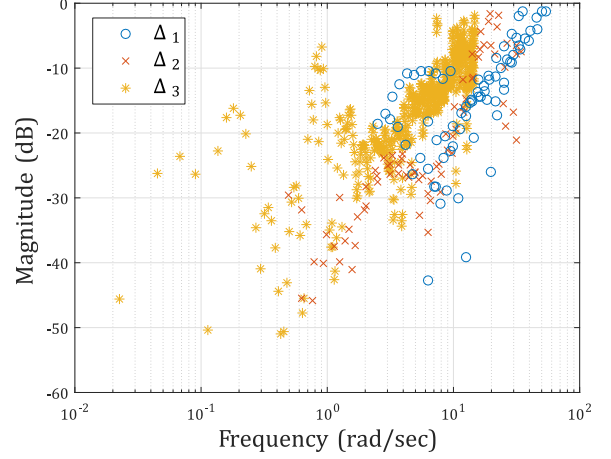


Fig. 7. Magnitude of uncertainty $|\Delta_i|$ from the three experimental identifications over various amplitude excitations. Δ_1 is motor-driven, no-load, Δ_2 , motor-driven, fixed-output, and Δ_3 is through manual excitation of output θ .

($\tau_m \rightarrow \phi$) as seen in (16), a direct identification of no-load motor dynamics. The second load condition fixes load position θ with the spring installed, with the dynamics seen in (17). The third identification keeps the spring and identifies the response from environmental input through prolonged, manually induced motion and no motor input. For the last test, the input θ and output ϕ are transformed to the frequency domain and divided to give system response as

$$\frac{\phi}{\tau_m} = \hat{G}(1 + \Delta_1) \quad (16)$$

$$\frac{\phi}{\tau_m} = \frac{K\hat{G}(1 + \Delta_2)}{1 + \alpha K\hat{G}(1 + \Delta_2)} \quad (17)$$

$$\frac{\phi}{\theta} = \frac{\alpha K\hat{G}(1 + \Delta_3)}{1 + \alpha K\hat{G}(1 + \Delta_3)} \quad (18)$$

The transfer functions for all these identifications (16)–(18) can then be inverted, using nominal model parameters (from earlier) and identified $\alpha = 0.62$ to find $\Delta_i^j \in \mathbb{C}$, the model deviation which explains the j th data point of the i th configuration. These uncertainties (Δ_i^j) capture the uncertainty arising from back-driveability as well as other effects (spring uncertainty, nonlinear friction, etc.), and their magnitudes are shown in Fig. 7. The uncertainty from the third identification is larger at lower frequencies, as each frequency input could not be isolated, allowing transient and nonlinear responses to be more significantly captured. For this reason, the uncertainties Δ_1 and Δ_2 will be used in the following analysis, and Δ_3 will only be used to confirm trends.

B. Design With Identified Uncertainty

The passivity relaxation frequency ω_0 is motivated from damping and inertia of the fixed load dynamics (e.g., inertial load and exoskeleton dynamics in Fig. 1), which are identified as $M_e = 0.31$, $B_e = 0.56$, and an upper bound on the environmental stiffness that can be expected, taken as $K_e =$

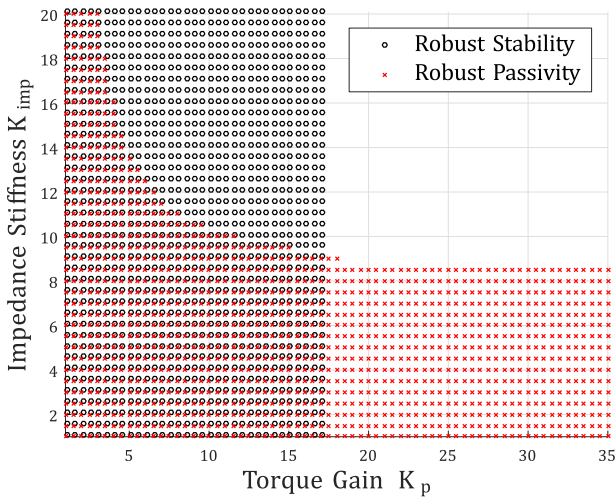


Fig. 8. Parameter bounds which meet robust passivity (relaxed with $\omega_0 = 60$ rad/s), and robust stability, where identified uncertainty is extrapolated to higher frequencies.

1000 N·m/rad. Bounding the controlled response gain by the physical stiffness K , the small gain condition is held above 60 rad/s.

Taking a fixed structure controller and the sets of observed deviations Δ_1 and Δ_2 , the robust passivity condition (6) can be used to motivate bounds on outer- and inner-loop controller parameters. The same controller architecture as in Section IV-A is used, a PD inner-loop torque controller and a stiffness and damping impedance controller. As the derivative term K_d uniformly improves passivity robustness, it is taken to its realizable limit $K_d \approx 1.5$, found experimentally. Impedance damping was fixed at $B_{\text{imp}} = 0.01$, which gave the maximum robustness in Table I and introduced acceptable amounts of noise in practice. With the damping terms fixed, the ranges of K_p and K_{imp} which meet robust passivity and stability can be found.

A relaxation for robust stability cannot be motivated and thus must hold across all frequencies. The observed uncertainty is extended to higher frequencies by fitting a 20 dB/dec line onto the experimentally found Δ_1 and Δ_2 , crossing 0 dB at 68 rad/s. The resulting bounds on K_d and K_{imp} can be seen in Fig. 8.

C. Passivity Violation

Although passivity cannot be validated on a system through finite input/output observations, violations of passivity can be observed by showing energy extraction over a trajectory closed in the state space. The input, which induced this trajectory, can be repeated to extract arbitrary amounts of energy. Two sets of parameters were realized, where $K_{\text{imp}} = 8$, $B_{\text{imp}} = 0.01$, and $K_p = 15$ for both, but inner-loop derivative action was low damping $K_d = 0.1$ and high damping $K_d = 1.5$. Both parameters satisfy nominal passivity (2), but low damping (K_d) does not satisfy robust passivity as seen in Fig. 9. The response of the system under these two parameter sets was then investigated by excitation of the load-side position, both manually and with impulses delivered by a dead blow hammer.

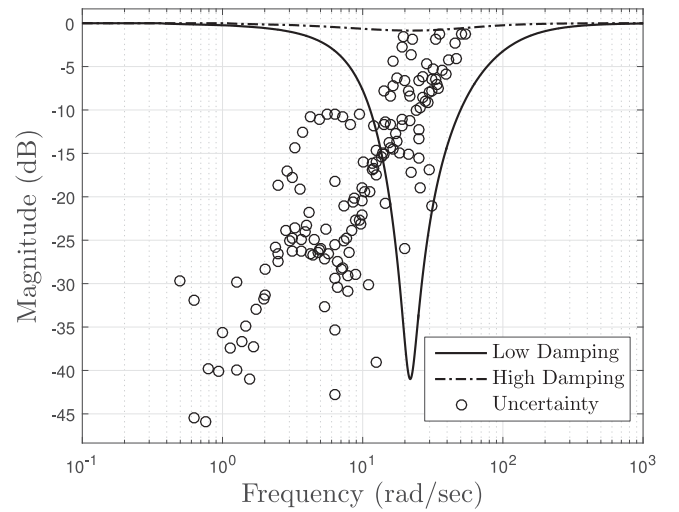


Fig. 9. Allowable uncertainty $\bar{\Delta}_p$ of the two sets of parameters displayed over the identified uncertainty Δ_1 and Δ_2 .

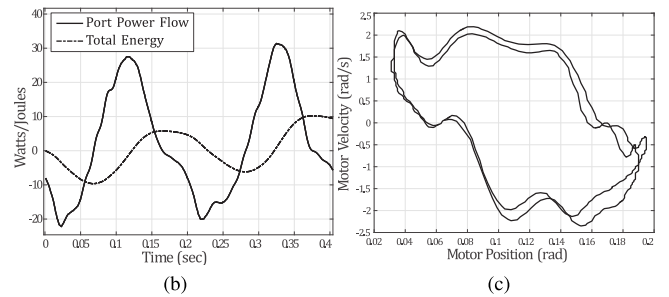
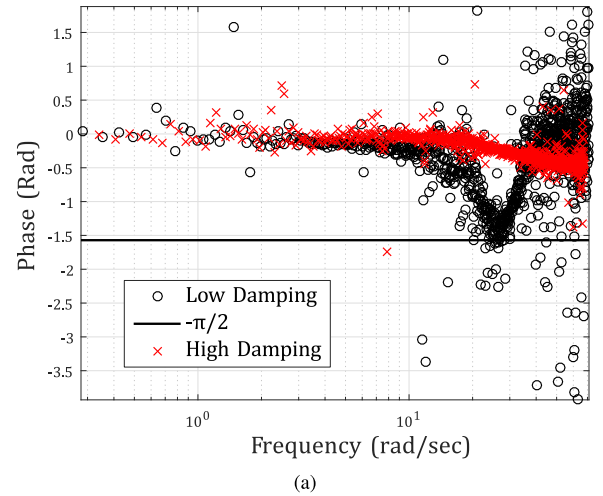


Fig. 10. (a) Phase response $\dot{\theta} \rightarrow \tau$ of the high and low damping systems. (b) and (c) Portion of the low damping system response under 28 rad/s manual excitation. As power is extracted (b) over a closed trajectory (c), the low damping system is not passive.

The phase response of these two systems can be seen in Fig. 10(a). The low damping system comes very close to a passivity violation near 28 rad/s. This was further examined by manual excitation near this frequency, and the port power flow [Fig. 10(b)] and state-space trajectory [Fig. 10(c)] induced during a portion of this input can be seen. As energy is extracted

over a trajectory which is closed in the state space, passivity is indeed violated.

D. Discussion

The model uncertainty shown in Fig. 7 captures not only the uncertainty arising from varying magnitudes of input, but also from the change between no-load (infinite environmental admittance) and fixed-output (zero environmental admittance) conditions. This may not generalize to intermediate load conditions, but suggests a reasonable bound. Model uncertainty under external excitation (Δ_3) is significantly higher, especially at lower frequencies. This could be due to the excitation not being at isolated frequencies, and also due to the lack of Coulomb friction compensation.

Applying robust passivity and stability with the identified uncertainty induces parameter constraints similar to those found experimentally. This suggests that robust passivity is not only useful for comparative design (i.e., the relative impact of a parameter or comparison of control architectures) but also in an absolute sense.

The resulting system response in Fig. 10 shows a violation of passivity for the low damping system occurring at the frequency predicted by Fig. 9 (28 rad/s). As the low damping system only met nominal passivity but not robust passivity, it can be seen that robust passivity improves the sufficiency of passivity in at least some cases. However, robust passivity, as developed here, is still not a universal safety condition, and higher bandwidth systems may also need to consider discretization effects.

VI. CONCLUSION

The fundamental tradeoff between performance and robustness remains relevant to interactive robots, requiring tools to find a suitable compromise. Toward understanding the safety limits of interactive robots analytically, this paper applied several techniques from robust control to impedance control. Extending passivity to hold over model uncertainty can make passivity more informative, allowing comparative studies on the effect of model and controller parameters. With a well-motivated uncertainty, it can also be used to find parameter bounds. Mixed passivity and small-gain coupled stability condition was introduced, which allows passivity to be violated at high frequencies, where typical environment and robot meet small-gain conditions.

Two major limitations of the approach are in the identification of uncertainty and the extension to high-DOF manipulators. Ideally, model and uncertainty identification can be done under the end use load conditions, but identifying the two extreme cases, zero and infinite environment admittance, can be a useful alternative. For high-DOF robots, identification under multiple load conditions may not be feasible if actuators have already been integrated. If an analytical form of uncertainty (e.g., parametric, friction, etc.) cannot be motivated, this approach cannot be used for controller synthesis. However, comparative studies can still help inform controller structure and identify the impact of fundamental parameters.

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